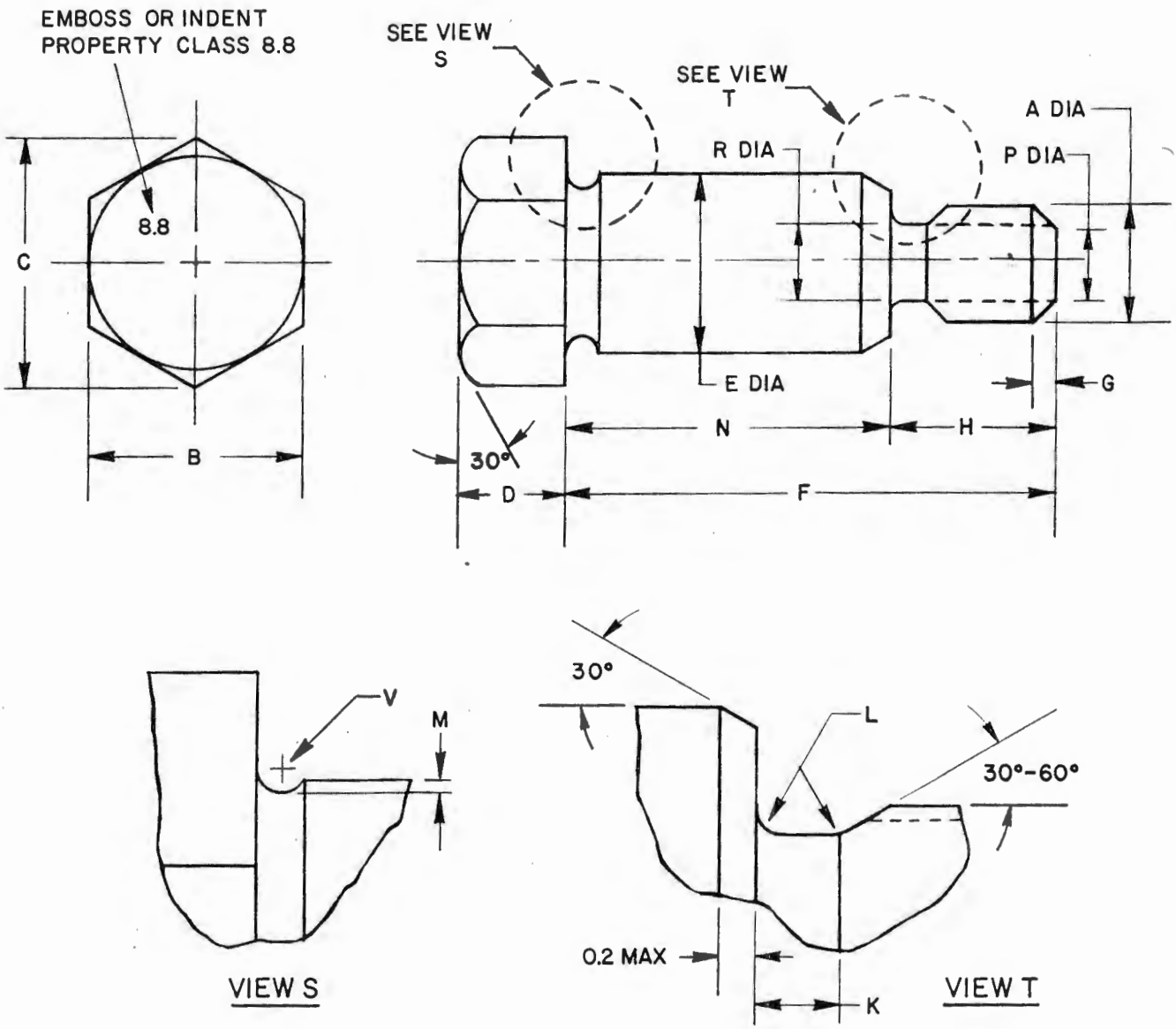
METRIC

MILITARY SPECIFICATION SHEET

BOLTS, SHOULDER, HEXAGON HEAD, SHORT THREAD, METRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the bolts described herein shall consist of this document and the latest issue of specification DOD-B-63475.



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FSC 5306

TABLE I. Dimensions (mm).

	A Thd dia & pitch	M10X1.5	M12X1.75	M14X2	M16X2	M18X2.5	M20X2.5	M22X2.5	M24X3	M27X3	M30X 3.5	M36X4
B	min	16.73	18.67	21.67	23.67	26.67	29.67	31.61	35.38	40.38	45.38	54.26
	max	17.00	19.00	22.00	24.00	27.00	30.00	32.00	36.00	41.00	46.00	55.00
C	min	18.90	21.10	24.49	26.75	30.14	33.53	35.72	39.98	45.63	51.28	61.31
D		7±0.18	8±0.18	9±0.18	10±0.18	12±0.215	13±0.215	14±0.215	15±0.215	17±0.215	19±0.26	23± 0.26
E	min	11.0	13.0	15.0	17.0	19.0	21.0	23.0	25.0	28.0	32.0	38.0
	max	11.2	13.2	15.2	17.2	19.2	21.3	23.3	25.3	28.3	32.3	38.3
F		SEE TABLE II										
G	max	2.2	2.5	3.0	3.0	3.5	3.5	3.5	4.5	4.5	5.0	6.0
H(ref)	1/	15.5	17.5	19.5	21.5	24.5	25.5	27.5	30.0	—	—	—
	2/	17.5	19.5	21.5	23.5	26.5	27.5	29.5	32.0	35.0	38.0	45.0
	3/	—	—	—	—	—	—	—	—	40.0	43.0	50.0
K	max	1.9	2.2	2.5	2.5	3.2	3.2	3.2	3.8	3.8	4.5	5.0
L	max	1.0	1.0	1.0	1.0	1.0	1.6	1.6	1.6	1.6	1.6	2.5
M	max	0.2	0.2	0.2	0.2	0.2	0.3	0.3	0.3	0.3	0.3	0.3
N		SEE TABLE II										
P	max	7.938	9.601	11.271	13.271	15.930	16.624	19.930	19.955	24.261	25.306	30.654
R	min	7.480	9.130	10.730	12.730	14.130	16.070	18.070	19.270	22.270	24.670	29.910
	max	7.700	9.400	11.000	13.000	14.400	16.400	18.400	19.600	22.600	25.000	30.300
V	max	0.75	0.9	1.0	1.0	1.25	1.25	1.25	1.5	1.5	1.75	2.0

1/ For lengths (F) up to 50 mm incl.
2/ For lengths (F) over 50 mm, up to 150 mm incl.
3/ For lengths (F) above 150 mm.

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TABLE II. Shoulder length - N.

Dim F	M10 N	M10 Dash no.	M12 N	M12 Dash no.	M14 N	M14 Dash no.	M16 N	M16 Dash no.	M18 N	M18 Dash no.	M20 N	M20 Dash no.	M22 N	M22 Dash no.	M24 N	M24 Dash no.	M27 N	M27 Dash no.	M30 N	M30 Dash no.	M36 N	M36 Dash no.
30	14.5	-001	12.5	-016																		
35	19.5	-002	17.5	-017	15.5	-033	13.5	-049														
40	24.5	-003	22.5	-018	20.5	-034	18.5	-050	15.5	-068	14.5	-086										
45	29.5	-004	27.5	-019	25.5	-035	23.5	-051	20.5	-069	19.5	-087	17.5	-104	15.0	-121						
50	34.5	-005	32.5	-020	30.5	-036	28.5	-052	25.5	-070	24.5	-088	22.5	-105	20.0	-122						
55	37.5	-006	35.5	-021	33.5	-037	31.5	-053	28.5	-071	27.5	-089	25.5	-106	23.0	-123	20.0	-138	17.0	-158		
60	42.5	-007	40.5	-022	38.5	-038	36.5	-054	33.5	-072	32.5	-090	30.5	-107	28.0	-124	25.0	-139	22.0	-159		
65	47.5	-008	45.5	-023	43.5	-039	41.5	-055	38.5	-073	37.5	-091	35.5	-108	33.0	-125	30.0	-140	27.0	-160	20.0	-178
70	52.5	-009	50.5	-024	48.5	-040	46.5	-056	43.5	-074	42.5	-092	40.5	-109	38.0	-126	35.0	-141	32.0	-161	25.0	-179
75	57.5	-010	55.5	-025	53.5	-041	51.5	-057	48.5	-075	47.5	-093	45.5	-110	43.0	-127	40.0	-142	37.0	-162	30.5	-180
80	62.5	-011	60.5	-026	58.5	-042	56.5	-058	53.5	-076	52.5	-094	50.5	-111	48.0	-128	45.0	-143	42.0	-163	35.0	-181
85	67.5	-012	65.5	-027	63.5	-043	61.5	-059	58.5	-077	57.5	-095	55.5	-112	53.0	-129	50.0	-144	47.0	-164	40.0	-182
90	72.5	-013	70.5	-028	68.5	-044	66.5	-060	63.5	-078	62.5	-096	60.5	-113	58.0	-130	55.0	-145	52.0	-165	45.0	-183
95	77.5	-014	75.5	-029	73.5	-045	71.5	-061	68.5	-079	67.5	-097	65.5	-114	63.0	-131	60.0	-146	57.0	-166	50.0	-184
100	82.5	-015	80.5	-030	78.5	-046	76.5	-062	73.5	-080	72.5	-098	70.5	-115	68.0	-132	65.0	-147	62.0	-167	55.0	-185
110			90.5	-031	88.5	-047	86.5	-063	83.5	-081	82.5	-099	80.5	-116	78.0	-133	75.0	-148	72.0	-168	65.0	-186
120			100.5	-032	98.5	-048	96.5	-064	93.5	-082	92.5	-100	90.5	-117	88.0	-134	85.0	-149	82.0	-169	75.0	-187
130							106.5	-065	103.5	-083	102.5	-101	100.5	-118	98.0	-135	95.0	-150	92.0	-170	85.0	-188
140							116.5	-066	113.5	-084	112.5	-102	110.5	-119	108.0	-136	105.0	-151	102.0	-171	95.0	-189
150							126.5	-067	123.5	-085	122.5	-103	120.5	-120	118.0	-137	115.0	-152	112.0	-172	105.0	-190
160																	120.0	-153	117.0	-173	110.0	-191
170																	130.0	-154	127.0	-174	120.0	-192
180																	140.0	-155	137.0	-175	130.0	-193
190																	150.0	-156	147.0	-176	140.0	-194
200																	160.0	-157	157.0	-177	150.0	-195

NOTES:

1. Dimensions. Unless otherwise specified, dimensions are in millimeters (mm) and are to be met after plating.
2. Part number. Consists of the letter "D" followed by the basic specification sheet number, a dash number taken from table II and suffix letter A or B. Example: D63475/2-001A. Suffix "A" indicates cadmium-plated and suffix "B" zinc-plated.

REQUIREMENTS:

1. Material. Alloy steel.
2. Protective finish. Cadmium plate or zinc plate (see note 2).
3. Tolerances. On length (F), ± 0.4 for sizes M10 thru M22, and ± 0.5 for sizes M24 thru M36. On shoulder length (N), ± 1.0.
4. Surface hardness. 20-30HRC.

Custodians:
Army - AR
Navy - SH
Air Force - 99

Review activities:
Army - AV, MI, AT
Navy - SH
Air Force - 11
DLA-IS
NSA-NS

User activities:
Army - CR, ME
Navy - AS, MC, OS

Preparing activity:
Army - AR

Agent:
DLA-IS

(Project 5306-0649)

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